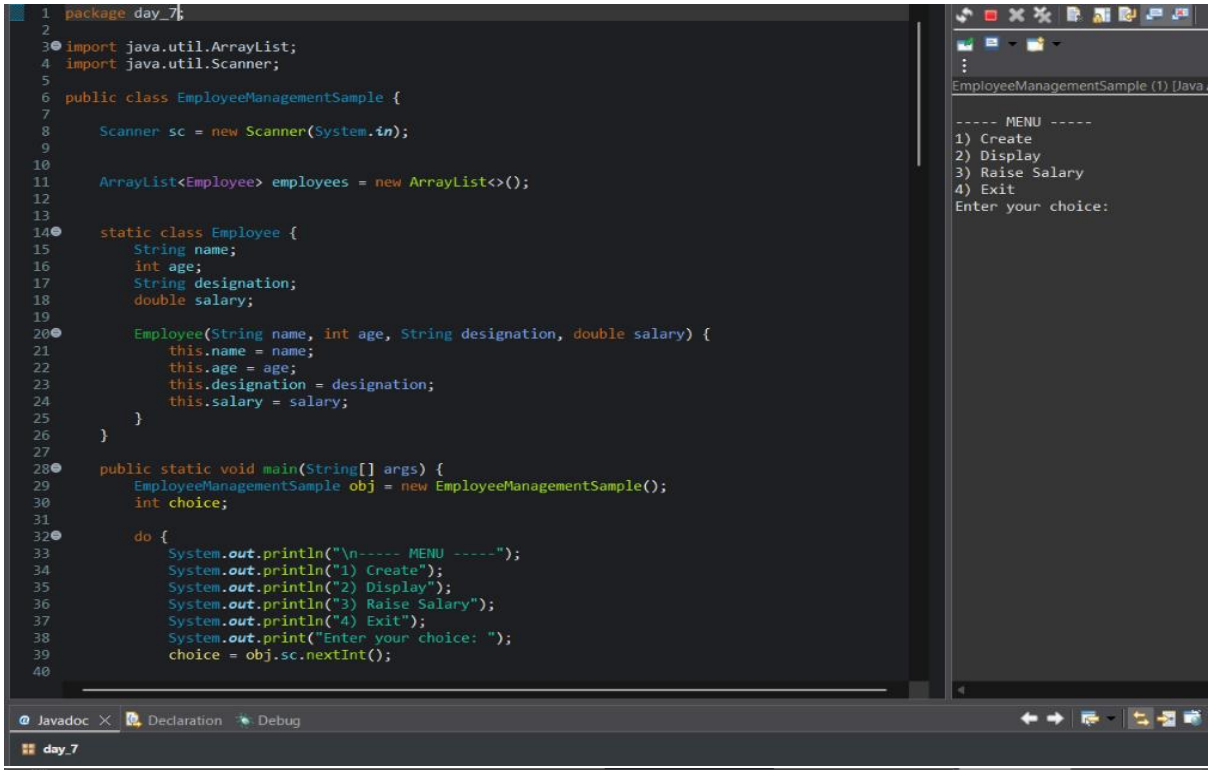
Project: Employee Management System (Console-based Java Application)

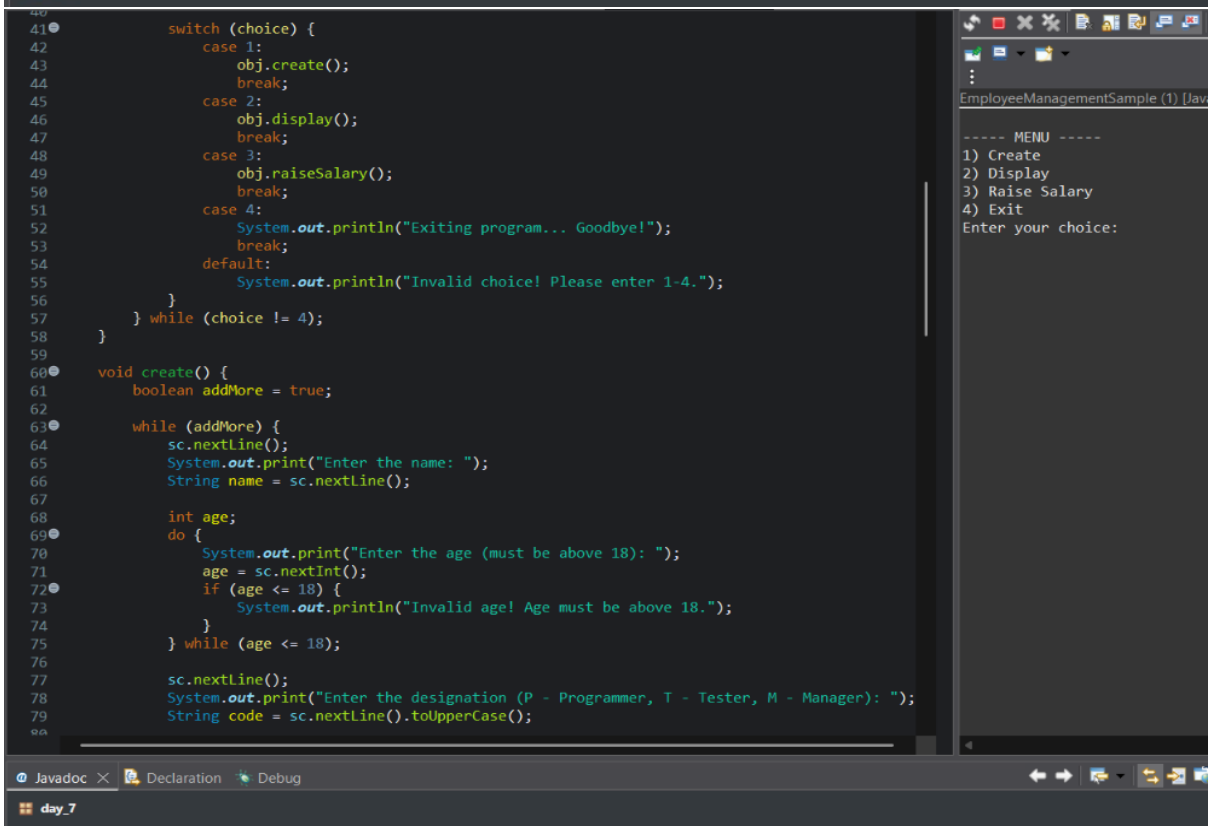
CODE:



```java
package day_7;

import java.util.ArrayList;
import java.util.Scanner;

public class EmployeeManagementSample {

    Scanner sc = new Scanner(System.in);


    ArrayList<Employee> employees = new ArrayList<>();


    static class Employee {
        String name;
        int age;
        String designation;
        double salary;

        Employee(String name, int age, String designation, double salary) {
            this.name = name;
            this.age = age;
            this.designation = designation;
            this.salary = salary;
        }
    }

    public static void main(String[] args) {
        EmployeeManagementSample obj = new EmployeeManagementSample();
        int choice;

        do {
            System.out.println("\n----- MENU -----");
            System.out.println("1) Create");
            System.out.println("2) Display");
            System.out.println("3) Raise Salary");
            System.out.println("4) Exit");
            System.out.print("Enter your choice: ");
            choice = obj.sc.nextInt();

```

```
EmployeeManagementSample (1) [Java

----- MENU -----
1) Create
2) Display
3) Raise Salary
4) Exit
Enter your choice:
```

```
Javadoc  Declaration  Debug
day_7
```



```java
            switch (choice) {
                case 1:
                    obj.create();
                    break;
                case 2:
                    obj.display();
                    break;
                case 3:
                    obj.raiseSalary();
                    break;
                case 4:
                    System.out.println("Exiting program... Goodbye!");
                    break;
                default:
                    System.out.println("Invalid choice! Please enter 1-4.");
            }
        } while (choice != 4);
    }

    void create() {
        boolean addMore = true;

        while (addMore) {
            sc.nextLine();
            System.out.print("Enter the name: ");
            String name = sc.nextLine();

            int age;
            do {
                System.out.print("Enter the age (must be above 18): ");
                age = sc.nextInt();
                if (age <= 18) {
                    System.out.println("Invalid age! Age must be above 18.");
                }
            } while (age <= 18);

            sc.nextLine();
            System.out.print("Enter the designation (P - Programmer, T - Tester, M - Manager): ");
            String code = sc.nextLine().toUpperCase();

```

```
EmployeeManagementSample (1) [Jav

----- MENU -----
1) Create
2) Display
3) Raise Salary
4) Exit
Enter your choice:
```

```
Javadoc  Declaration  Debug
day_7
```

```
            String designation;
            double salary;

            switch (code) {
                case "P":
                    designation = "Programmer";
                    salary = 35000;
                    break;
                case "T":
                    designation = "Tester";
                    salary = 25000;
                    break;
                case "M":
                    designation = "Manager";
                    salary = 50000;
                    break;
                default:
                    designation = "Not Specified";
                    salary = 0;
            }


            employees.add(new Employee(name, age, designation, salary));

            System.out.println("Employee created successfully! Base salary set to " + salary);


            String response;
            while (true) {
                System.out.print("Do you want to add another employee? (Y/N): ");
                response = sc.nextLine().trim().toUpperCase();
                if (response.equals("Y") || response.equals("N")) {
                    break;
                }
                System.out.println("Invalid input! Please enter Y or N.");
            }

            addMore = response.equals("Y");
        }
    }
```

```
EmployeeManagementSample (1) [Ja

----- MENU -----
1) Create
2) Display
3) Raise Salary
4) Exit
Enter your choice:
```

Javadoc | Declaration | Debug

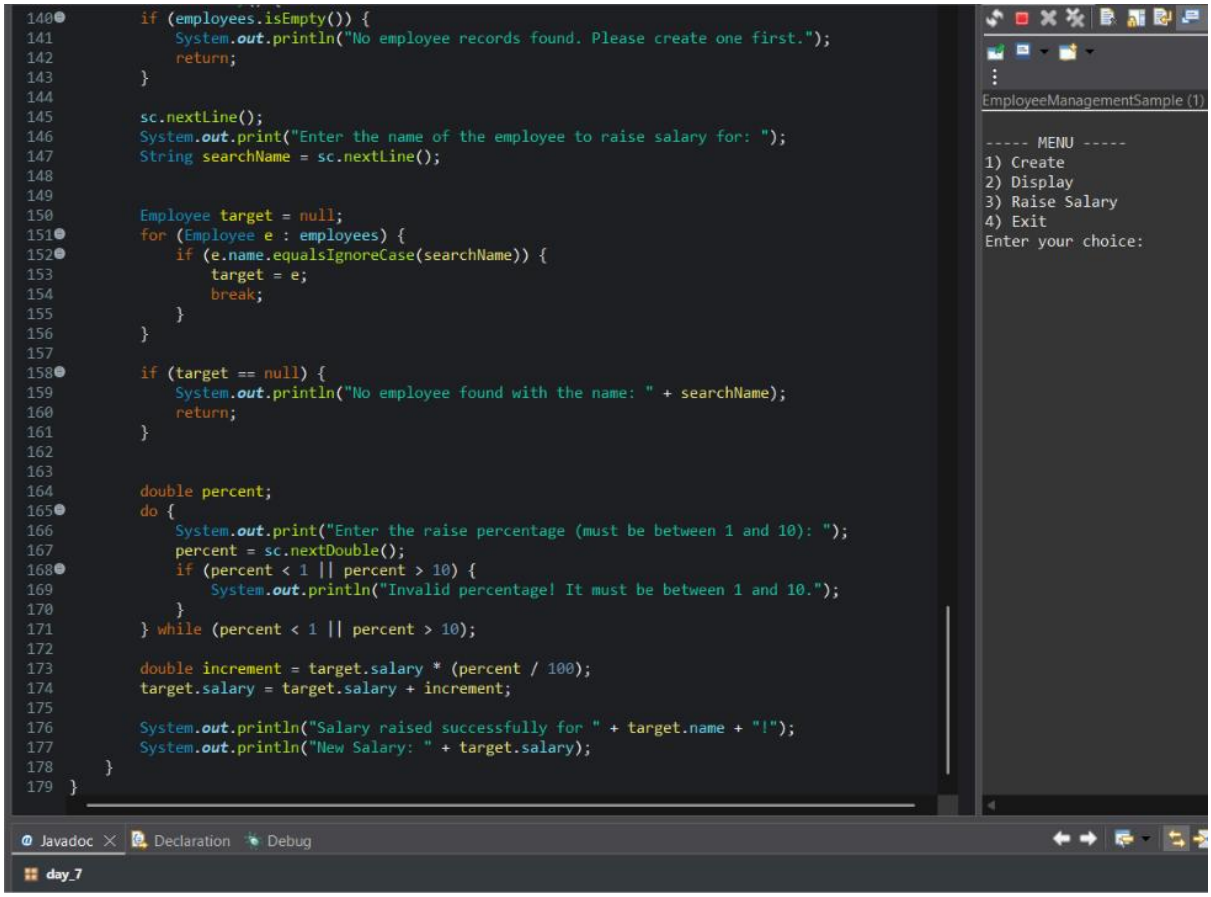
day_7

```
    void display() {
        if (employees.isEmpty()) {
            System.out.println("No employee records found. Please create one first.");
            return;
        }

        System.out.println("\n--- Employee Details ---");
        for (int i = 0; i < employees.size(); i++) {
            Employee e = employees.get(i);
            System.out.println("\nEmployee #" + (i + 1));
            System.out.println("Name: " + e.name);
            System.out.println("Age: " + e.age);
            System.out.println("Designation: " + e.designation);
            System.out.println("Salary: " + e.salary);
        }
    }

    void raiseSalary() {
        if (employees.isEmpty()) {
            System.out.println("No employee records found. Please create one first.");
            return;
        }

        sc.nextLine();
        System.out.print("Enter the name of the employee to raise salary for: ");
        String searchName = sc.nextLine();


        Employee target = null;
        for (Employee e : employees) {
            if (e.name.equalsIgnoreCase(searchName)) {
                target = e;
                break;
            }
        }

        if (target == null) {
            System.out.println("No employee found with the name: " + searchName);
            return;
        }
```

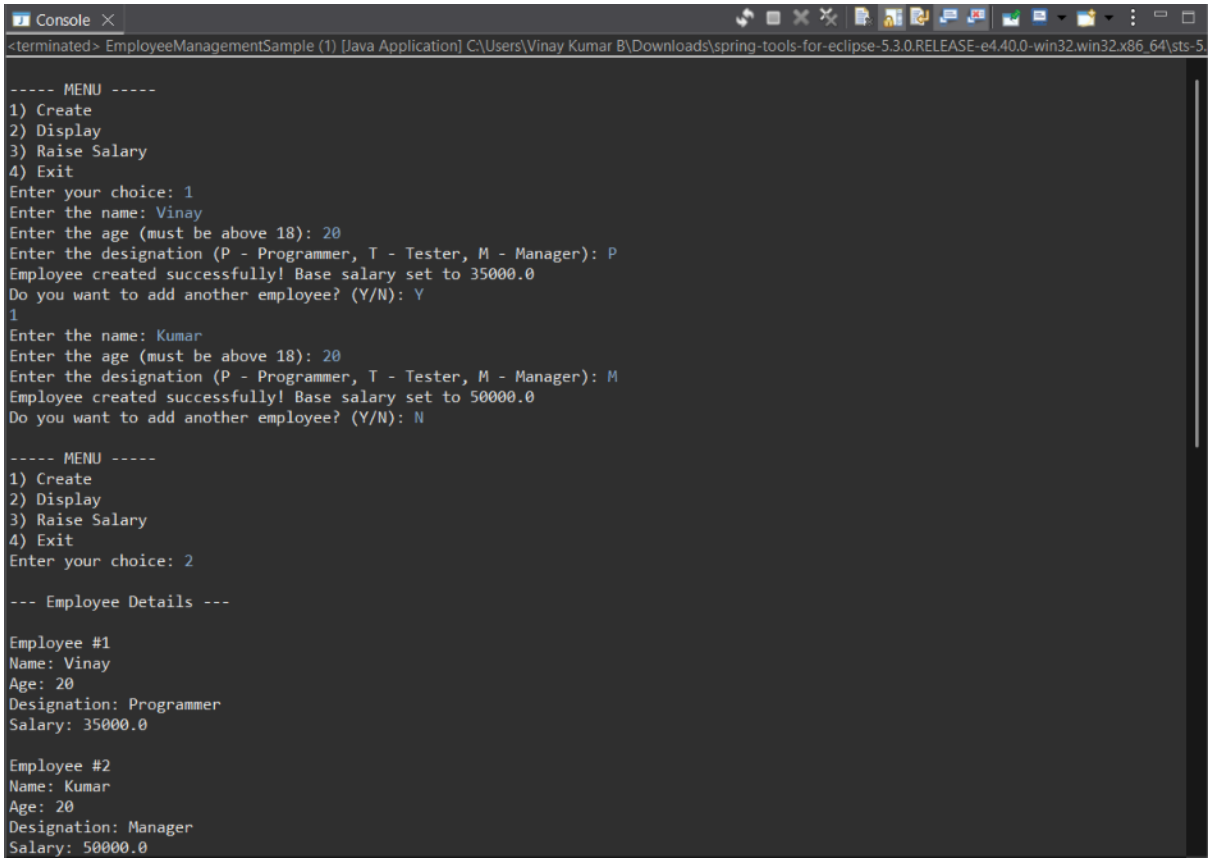
```
EmployeeManagementSamp

----- MENU -----
1) Create
2) Display
3) Raise Salary
4) Exit
Enter your choice:
```

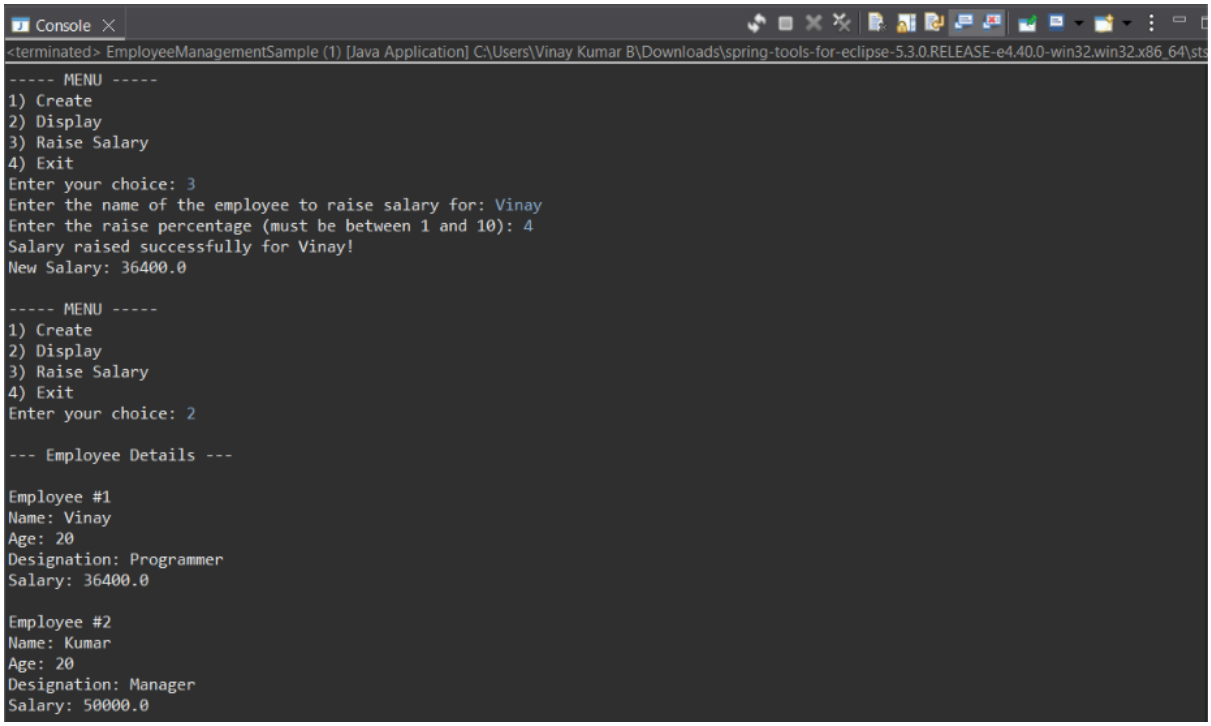
Javadoc | Declaration | Debug

day_7


if (employees.isEmpty()) {
System.out.println("No employee records found. Please create one first.");
return;
}
sc.nextLine();
System.out.print("Enter the name of the employee to raise salary for: ");
String searchName = sc.nextLine();
Employee target = null;
for (Employee e : employees) {
if (e.name.equalsIgnoreCase(searchName)) {
target = e;
break;
}
}
if (target == null) {
System.out.println("No employee found with the name: " + searchName);
return;
}
double percent;
do {
System.out.print("Enter the raise percentage (must be between 1 and 10): ");
percent = sc.nextDouble();
if (percent < 1 || percent > 10) {
System.out.println("Invalid percentage! It must be between 1 and 10.");
}
} while (percent < 1 || percent > 10);
double increment = target.salary * (percent / 100);
target.salary = target.salary + increment;
System.out.println("Salary raised successfully for " + target.name + "!");
System.out.println("New Salary: " + target.salary);
}
}
EmployeeManagementSample (1)
----- MENU -----
1) Create
2) Display
3) Raise Salary
4) Exit
Enter your choice:
Javadoc
Declaration
Debug
day_7

OUTPUT:


Console
<terminated> EmployeeManagementSample (1) [Java Application] C:\Users\Vinay Kumar B\Downloads\spring-tools-for-eclipse-5.3.0.RELEASE-e4.40.0-win32.win32.x86_64\sts-5.
----- MENU -----
1) Create
2) Display
3) Raise Salary
4) Exit
Enter your choice: 1
Enter the name: Vinay
Enter the age (must be above 18): 20
Enter the designation (P - Programmer, T - Tester, M - Manager): P
Employee created successfully! Base salary set to 35000.0
Do you want to add another employee? (Y/N): Y
1
Enter the name: Kumar
Enter the age (must be above 18): 20
Enter the designation (P - Programmer, T - Tester, M - Manager): M
Employee created successfully! Base salary set to 50000.0
Do you want to add another employee? (Y/N): N
----- MENU -----
1) Create
2) Display
3) Raise Salary
4) Exit
Enter your choice: 2
--- Employee Details ---
Employee #1
Name: Vinay
Age: 20
Designation: Programmer
Salary: 35000.0
Employee #2
Name: Kumar
Age: 20
Designation: Manager
Salary: 50000.0


Console
<terminated> EmployeeManagementSample (1) [Java Application] C:\Users\Vinay Kumar B\Downloads\spring-tools-for-eclipse-5.3.0.RELEASE-e4.40.0-win32.win32.x86_64\sts
----- MENU -----
1) Create
2) Display
3) Raise Salary
4) Exit
Enter your choice: 3
Enter the name of the employee to raise salary for: Vinay
Enter the raise percentage (must be between 1 and 10): 4
Salary raised successfully for Vinay!
New Salary: 36400.0
----- MENU -----
1) Create
2) Display
3) Raise Salary
4) Exit
Enter your choice: 2
--- Employee Details ---
Employee #1
Name: Vinay
Age: 20
Designation: Programmer
Salary: 36400.0
Employee #2
Name: Kumar
Age: 20
Designation: Manager
Salary: 50000.0


----- MENU -----
1) Create
2) Display
3) Raise Salary
4) Exit
Enter your choice: 4
Exiting program... Goodbye!
Declaration
Debug

What it does:

This is a simple menu-driven Java console application that lets a user manage employee records in memory using an ArrayList. It demonstrates core Java concepts like OOP, collections, loops, switch-case, and input validation.

Key Features:

1. **Create Employee :**
   - Takes employee name, age (validated to be above 18), and designation code (P/T/M).
   - Automatically assigns designation and base salary based on the code:
     - P → Programmer, ₹35,000
     - T → Tester, ₹25,000
     - M → Manager, ₹50,000
     - Anything else → "Not Specified", ₹0
   - Allows adding multiple employees in a loop (asks Y/N after each entry).

2. **Display Employees :**
   - Lists all stored employees with their name, age, designation, and salary.
   - Shows a message if no records exist.

3. **Raise Salary :**
   - Searches for an employee by name (case-insensitive).
   - Validates and applies a percentage-based salary raise (must be between 1% and 10%).
   - Updates and displays the new salary**.**
   -

4. **Exit :**
   - Ends the program gracefully.